Nikolay Kurtov

Senior Software Engineer | System Cryptography | Distributed Systems

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Summary

Seasoned Software Engineer with 15+ years of experience specializing in high-performance computing, large-scale distributed systems, and hardware acceleration (FPGA). Proven track record of leading technical initiatives at Google and NEAR Protocol, currently building the next generation of decentralized infrastructure at SubZero Labs (Rialo).

Experience

Senior Software Engineer, REX Team — SubZero Labs (July 2025 – Present)

- **Rialo Infrastructure:** Contributing to the core architecture of **Rialo**, a full-stack decentralized network designed for real-world applications and internet-scale performance.
- **REX (Runtime/Execution):** Part of the REX team focused on high-throughput, event-driven execution and **SVM compatibility** (proprietary codebase).
- **System Design:** Designing and implementing performance-critical components for a high-concurrency, low-latency consensus and execution environment.

System SW / Cryptography Engineer — Irreducible.com (2024 – June 2025)

- **Xlang Compiler:** Architected and implemented **Xlang**, a custom IR compiler for proprietary hardware cores. Achieved **99.5% hardware utilization** through advanced software pipelining and instruction scheduling.
- **FPGA Acceleration:** Delivered a **12x performance boost** for **Binius Sumchecks** and Zerochecks by offloading critical bottlenecks to FPGA (open-source link).
- **Team Leadership:** Mentored junior engineers to exploit fine-grained parallelism in Evalcheck, resulting in a **15x performance improvement**.
- **Production Deployment:** Designed the end-to-end software stack bridging high-level cryptographic protocols (Binius) with low-level FPGA drivers.

Tech Lead, Node Team — NEAR Protocol (2021 – 2023)

- **State Sync Re-architecture:** Led the complete redesign of the **State Sync mechanism**, significantly improving protocol reliability and node synchronization speed (design document).
- **Operational Excellence:** Reduced release-blocking incidents and improved the developer/validator experience by enhancing node monitoring with advanced metrics, structured logging, and **distributed tracing** (Grafana Tempo).
- **Core Contributions:** Maintained and optimized core blockchain protocol logic in **Rust**.

Senior Software Engineer — Google (2011 – 2021)

- **Tech Lead, YouTube Core SRE (2.5 years):** Directed end-to-end reliability for YouTube's global infrastructure. Led the development of a C++ microservice dependency framework and established production readiness standards for core services.
- **Knowledge Graph Infrastructure & Quality (4 years):**

  - Optimized backend components for massive-scale graph data, significantly increasing throughput.
  - Improved ML models for music entity reconciliation, impacting **0.3% of global Google Search traffic**.
- **Data Quality & Personalized Offers (3.5 years):** Developed ML-driven systems for deduplicating business listings and built high-scale personalized offer distribution systems (Email/Web).

Early Career (2006 – 2011)

- **Intel (2008 – 2011):** Software Engineer / Intern. Optimized C++ compilers for Atom processors and developed distributed memory runtimes for Intel Concurrent Collections.
- **Intel Student Lab / xored / SWSoft (2006 – 2008):** Contributions to Apache Harmony compiler (edge profiling), Debian systems administration, and 3D game engine development.

Skills

- **Languages:** Rust (4+ yrs), C++ (14+ yrs), Python, Go
- **Expertise:** High-Performance Computing (HPC), FPGA Acceleration, Distributed Systems, Blockchain Protocols, SRE & Scalability
- **Core Competencies:** System Architecture, Parallel Programming, Low-level Optimization, Cryptographic Engineering

Education

- **MS in Computer Science**, Novosibirsk State University (2009 – 2011). GPA: 5.0/5.0
- **BS in Computer Science**, Novosibirsk State University (2005 – 2009). GPA: 5.0/5.0

Programming Competitions

- **ACM ICPC World Finals:** 14th place (Harbin, 2010), 27th place (Orlando, 2011)
- **Yandex Open:** 11th place (Onsite round, 2010)

Hobbies

- **HEMA (Swordfighting):** Assistant Coach. Ranking
- **Running:** Marathon Personal Best: 3h 25m
- **Astronomy:** Built a DIY Solar Projection Telescope